

|                                        |            |             |
|----------------------------------------|------------|-------------|
| Name: Aranza Lucía Jiménez Orellana    |            |             |
| School: Academica Británica Cuscatleca |            |             |
| Group: 5th kinni...                    |            | Sex: Female |
| Date of first test: 14/11/2024         | Level: 9B  | Age: 9:08   |
| Date of second test: 13/10/2025        | Level: 10B | Age: 10:07  |

Scores

| No. attempted (/58) | SAS | SAS (with 90% confidence bands) |    |    |    |                        |     |     |     |     |  | Overall ST | PR | GR (/23) | End of KS2 indicator | Progress Category |
|---------------------|-----|---------------------------------|----|----|----|------------------------|-----|-----|-----|-----|--|------------|----|----------|----------------------|-------------------|
|                     |     | 60                              | 70 | 80 | 90 | 100                    | 110 | 120 | 130 | 140 |  |            |    |          |                      |                   |
| 55                  | 100 |                                 |    |    |    | <div><div></div></div> |     |     |     |     |  | 5          | 50 | 6        | 104                  | Much higher       |

Previous & Current Scores

| Date       | Age at test<br>(yrs:mths) | Level /<br>Form | No.<br>attempted | SAS | SAS (with 90% confidence bands) |    |    |    |     |     |     |     |     |  | ST | PR | End of<br>KS2<br>indicator |
|------------|---------------------------|-----------------|------------------|-----|---------------------------------|----|----|----|-----|-----|-----|-----|-----|--|----|----|----------------------------|
|            |                           |                 |                  |     | 60                              | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |  |    |    |                            |
| 08/05/2025 | 10:01                     | 10A             | 54 / 58          | 86  |                                 |    |    |    |     |     |     |     |     |  | 3  | 18 | 96                         |
| 13/10/2025 | 10:07                     | 10B             | 55 / 58          | 100 |                                 |    |    |    |     |     |     |     |     |  | 5  | 50 | 104                        |

Analysis of Curriculum content categories

| Curriculum content category | Number of questions | Student % correct | Standardisation % correct | Student / standardisation difference |
|-----------------------------|---------------------|-------------------|---------------------------|--------------------------------------|
| Number                      | 34                  | 50%               | 54%                       | -4%                                  |
| Measurement                 | 10                  | 27%               | 36%                       | -9%                                  |
| Geometry                    | 6                   | 57%               | 51%                       | 6%                                   |
| Statistics                  | 8                   | 100%              | 66%                       | 34%                                  |

Analysis of Process categories

| Process category                    | Number of questions | Student % correct | Standardisation % correct | Student / standardisation difference |
|-------------------------------------|---------------------|-------------------|---------------------------|--------------------------------------|
| Fluency in facts and procedures     | 9                   | 78%               | 63%                       | 15%                                  |
| Fluency in conceptual understanding | 22                  | 42%               | 52%                       | -10%                                 |
| Problem solving                     | 5                   | 50%               | 43%                       | 7%                                   |
| Mathematical reasoning              | 22                  | 56%               | 50%                       | 6%                                   |

## Implications for teaching and learning

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- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
  - Aranza Lucía took *PTM9B* in November 2024 and from then until now has made much higher than expected progress in maths.
- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- Where performance is fairly evenly balanced across curriculum categories, this suggests that Aranza Lucía will generally demonstrate a level of understanding of mathematical concepts appropriate for this age, irrespective of whether this is in a format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Fluency and agility are equally well developed in both written and mental formats and Aranza Lucía is developing the language of mathematics in line with expectations for her age group.
- Where performance across the curriculum categories are uneven, specific areas of weakness might be addressed as follows:
  - Further targeted practice in the areas identified as being relatively weaker.
  - Practical activities using equipment that is designed to help Aranza Lucía to 'see' the thinking that lies behind any concepts that are not yet secure.
  - Get Aranza Lucía to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Aranza Lucía is generally secure in performing the basic mental calculations expected for this age group. These include age-appropriate fluency with whole numbers, fractions, decimals, percentages and ratio as well as an understanding of the number system, place value (in 4 and 5 digit numbers) and the connections between inverse operations such as multiplication and division.
- Aranza Lucía is developing mathematical reasoning and the ability to solve a wider range of problems, including increasingly complex properties of numbers and arithmetic, and problems demanding efficient written and mental methods of calculation. Aranza Lucía has been introduced to the language of algebra to extend problem solving to a wider variety of contexts and is also developing an understanding of the geometry of increasingly complex shapes and the language needed to describe these.
- Next steps should include opportunities to stretch and develop conceptual understanding, to increase mental agility and to progress the development of formal written mathematics using age-appropriate symbols and methodologies. This can be done by emphasising the higher order skills of hypothesising or predicting (What is an approximate value for  $49 \times 11$ ?); designing and comparing procedures (How many different ways can we work out  $53 - 18$ ? Which is the most efficient?); interpreting results (If  $9 \times 5 = 45$ , how many other calculations can we easily work out, e.g.  $0.9 \times 5$ ?) and applying reasoning (If I have £7.39 in my purse, what coins might they be?).
- In the formal written development there should be an emphasis on developing the correct use of mathematical language, for example in using ':' for ratio and the '=' sign correctly as a mathematical sentence.
- By providing practice time and frequent opportunities for Aranza Lucía to use one or more known facts to work out more facts, and by engaging Aranza Lucía in discussion and providing opportunities to explain methods and strategies to you and her peers, Aranza Lucía will be helped to make the connections between mental calculations and the written explanations. This will help to develop both problem solving abilities and the language of mathematics, thus improving both mental agility and the ability to plan and carry out accurate calculations in a more formal way.

|                                               |                   |                    |
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## Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

| Question number | Curriculum category | Process category                    | Question content                                                                                                           | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| MM1             | Number              | Fluency in facts and procedures     | Write in figures the number four hundred and thirty-seven.                                                                 | 1 / 1      | 91              | 90                        |
| MM2             | Number              | Fluency in facts and procedures     | What number is halfway between three point eight and five point eight?                                                     | 1 / 1      | 78              | 73                        |
| MM3             | Number              | Fluency in conceptual understanding | Write one quarter as a percentage.                                                                                         | 0 / 1      | 35              | 54                        |
| MM4             | Number              | Mathematical reasoning              | Write in figures five hundred and seventy-five correct to the nearest hundred.                                             | 1 / 1      | 83              | 77                        |
| MM5             | Number              | Fluency in conceptual understanding | Add two point two three and one point zero one.                                                                            | 1 / 1      | 70              | 72                        |
| MM6             | Number              | Fluency in conceptual understanding | There are thirty-one days in January. After the seventeenth of January, how many days are left until the end of the month? | 0 / 1      | 39              | 50                        |
| MM7             | Measures            | Mathematical reasoning              | How many weeks are forty-nine days?                                                                                        | 1 / 1      | 26              | 51                        |
| MM8             | Number              | Fluency in conceptual understanding | What is forty-nine multiplied by a hundred?                                                                                | 0 / 1      | 48              | 48                        |
| MM9             | Number              | Fluency in conceptual understanding | I have weeded seventy percent of my path. What percentage is left to weed?                                                 | 1 / 1      | 87              | 89                        |
| MM10            | Number              | Fluency in facts and procedures     | Tom has sixteen identical socks. How many pairs of socks does he have?                                                     | 1 / 1      | 61              | 70                        |
| MM11            | Measures            | Fluency in facts and procedures     | Bob spends three pounds seventy pence. He pays with a five-pound note. How much change does he get?                        | 0 / 1      | 39              | 49                        |
| MM12            | Number              | Fluency in conceptual understanding | The temperature was minus three degrees Celcius. It went up by ten degrees. Write the new temperature.                     | 0 / 1      | 65              | 69                        |
| MM13            | Number              | Mathematical reasoning              | Forty-eight strawberries are shared between six people. How many does each person get?                                     | 1 / 1      | 48              | 67                        |
| MM14            | Number              | Fluency in conceptual understanding | One fifth of a number is four. What is the number?                                                                         | 0 / 1      | 22              | 52                        |
| MM15            | Measures            | Fluency in facts and procedures     | A TV programme starts at ten past five and ends at ten past seven. How many minutes does the programme last?               | 0 / 1      | 13              | 45                        |
| AU1             | Number              | Fluency in conceptual understanding | How far did he run in total on Tuesday and Thursday?                                                                       | 1 / 1      | 39              | 67                        |
| AU2             | Number              | Fluency in conceptual understanding | How much further did he run on Saturday than on Wednesday?                                                                 | 0 / 1      | 17              | 41                        |

| Question number | Curriculum category | Process category                    | Question content                                                                                | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|-------------------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| AU3             | Number              | Fluency in conceptual understanding | How far did David run on Wednesday?                                                             | 0 / 1      | 39              | 27                        |
| AU4             | Statistics          | Fluency in conceptual understanding | Which is the youngest cat?                                                                      | 1 / 1      | 87              | 83                        |
| AU5             | Statistics          | Mathematical reasoning              | What is the difference in mass between the lightest cat and the heaviest cat?                   | 1 / 1      | 35              | 46                        |
| AU6             | Statistics          | Problem solving                     | Write the labels in the correct spaces.                                                         | 1 / 1      | 96              | 82                        |
| AU7a            | Number              | Problem solving                     | How many packs of books does Miss Begum need to buy?                                            | 1 / 1      | 74              | 62                        |
| AU7b            | Number              | Fluency in conceptual understanding | How much will she spend?                                                                        | 0 / 1      | 48              | 39                        |
| AU8             | Statistics          | Fluency in facts and procedures     | How many pencils, pieces of paper and pens does she need for 10 pupils?                         | 1 / 1      | 87              | 76                        |
| AU9             | Statistics          | Mathematical reasoning              | Write the labels in the spaces on the correct lines.                                            | 1 / 1      | 87              | 78                        |
| AU10            | Number              | Fluency in conceptual understanding | How much shorter do they need to be?                                                            | 0 / 1      | 13              | 40                        |
| AU11            | Measures            | Mathematical reasoning              | Joe buys 4 bananas. What is the smallest amount they can weigh altogether?                      | 0 / 1      | 30              | 41                        |
| AU12            | Measures            | Mathematical reasoning              | Joe buys 1 banana, 10 grapes and 2 plums. What is the largest amount they can weigh altogether? | 0 / 1      | 35              | 27                        |
| AU13            | Measures            | Problem solving                     | Joe buys 500 grams of grapes. What is the largest number of grapes he can get?                  | 0 / 1      | 13              | 23                        |
| AU14a           | Number              | Fluency in conceptual understanding | Circle the name of the city that is warmest in winter.                                          | 1 / 1      | 87              | 77                        |
| AU14b           | Number              | Mathematical reasoning              | Cardiff is 8 degrees warmer than Oslo. What is the temperature in Cardiff?                      | 0 / 1      | 48              | 61                        |
| AU15            | Measures            | Fluency in conceptual understanding | How many cubes does he use? Use the cards to show how to work it out. Write your answer         | 2 / 2      | 33              | 33                        |
| AU16a           | Number              | Fluency in conceptual understanding | Which is the odd one out?                                                                       | 1 / 1      | 35              | 59                        |
| AU16b           | Number              | Mathematical reasoning              | Write another fraction with the same value as the other 4 fractions.                            | 0 / 1      | 35              | 38                        |
| AU17            | Number              | Problem solving                     | How many children are there at the camp?                                                        | 0 / 2      | 9               | 22                        |
| AU18a           | Number              | Mathematical reasoning              | Which rod is two-thirds of rod B?                                                               | 0 / 1      | 13              | 38                        |
| AU18b           | Number              | Mathematical reasoning              | Which rod is 50 per cent of rod E?                                                              | 1 / 1      | 48              | 57                        |
| AU18c           | Number              | Mathematical reasoning              | Fill in the fraction.                                                                           | 1 / 1      | 17              | 25                        |
| AU19a           | Number              | Fluency in conceptual understanding | Write these fractions in the correct boxes.                                                     | 1 / 2      | 46              | 61                        |
| AU19b           | Number              | Mathematical reasoning              | Make up one more fraction for each box.                                                         | 1 / 1      | 26              | 55                        |
| AU20            | Number              | Problem solving                     | How many adults are needed on the trip?                                                         | 1 / 1      | 39              | 48                        |
| AU21            | Geometry            | Mathematical reasoning              | Circle all the squares that are cut into 2 equal pieces.                                        | 1 / 2      | 52              | 56                        |
| AU22            | Geometry            | Mathematical reasoning              | Circle all the squares where the cut is a line of symmetry.                                     | 0 / 1      | 26              | 40                        |

| Question number | Curriculum category | Process category                    | Question content                                                                        | Score (/x) | Group % correct | Standardisation % correct |
|-----------------|---------------------|-------------------------------------|-----------------------------------------------------------------------------------------|------------|-----------------|---------------------------|
| AU23            | Number              | Fluency in conceptual understanding | Work out the other ways and fill in the gaps below.                                     | 0 / 2      | 17              | 18                        |
| AU24            | Number              | Mathematical reasoning              | How many boxes do they need to buy? How many bags of crisps will be left over?          | 1 / 1      | 39              | 44                        |
| AU25            | Number              | Mathematical reasoning              | Put the correct numbers in the spaces.                                                  | 1 / 3      | 48              | 57                        |
| AU26            | Measures            | Fluency in conceptual understanding | What is the perimeter of her vegetable garden?                                          | 0 / 1      | 17              | 36                        |
| AU27a           | Measures            | Mathematical reasoning              | Which diagram shows a vegetable garden that has the same area, but a smaller perimeter? | 0 / 1      | 17              | 33                        |
| AU27b           | Measures            | Fluency in conceptual understanding | What is the perimeter of the new vegetable garden?                                      | 0 / 1      | 17              | 29                        |
| AU28a           | Number              | Fluency in conceptual understanding | Sort the cities in order of distance from London.                                       | 2 / 2      | 70              | 71                        |
| AU28b           | Number              | Mathematical reasoning              | How much further away from London is Honolulu than Miami?                               | 0 / 1      | 30              | 39                        |
| AU29            | Geometry            | Fluency in facts and procedures     | Which is the smallest angle?                                                            | 1 / 1      | 43              | 66                        |
| AU30            | Geometry            | Fluency in facts and procedures     | Which is the largest angle?                                                             | 1 / 1      | 39              | 52                        |
| AU31            | Geometry            | Fluency in facts and procedures     | Which 2 angles are reflex angles?                                                       | 1 / 1      | 48              | 46                        |
| AU32            | Geometry            | Fluency in conceptual understanding | Which 2 angles together would make 1 complete turn?                                     | 0 / 1      | 52              | 41                        |
| AU33i           | Statistics          | Mathematical reasoning              | Draw the correct scores on the bar chart.                                               | 1 / 1      | 91              | 71                        |
| AU33ii          | Statistics          | Mathematical reasoning              | Draw the correct scores on the bar chart.                                               | 1 / 1      | 35              | 53                        |
| AU33iii         | Statistics          | Mathematical reasoning              | Draw the correct scores on the bar chart.                                               | 1 / 1      | 35              | 38                        |